Background Shortcut Learning Mitigation for Waterbirds

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Abstract. This is our weekly report on three methods that try to stop image classifiers from relying on the background: DPE, POTER, and AutoBackSwap. We ran each method on the Waterbirds dataset and compared our numbers with the ones in the papers. DPE and AutoBackSwap came close to the published worst-group accuracy. POTER finished training for one seed, but we only have validation results so far. While looking at AutoBackSwap's mistakes, we found that three bird species seem to be labeled wrongly in the dataset, and that they cause many of its errors. We also list three ideas for improving DPE that we plan to try next week.

Keywords: Shortcut learning · Waterbirds · Reproducibility · Worst-group accuracy.

1 Introduction

1.1 Problem statement

Definition. Each image in the Waterbirds dataset [4] shows a bird, and the task is to decide whether it is a landbird or a waterbird. The difficulty is that the background is correlated with the label: most landbirds are photographed on land and most waterbirds on water. A model can therefore get good accuracy just by looking at the background, without looking at the bird.

Why it is hard. A normally trained model (ERM, which simply minimizes the average training loss) tends to learn this shortcut. It then fails on the rare images where the background is misleading, such as a landbird on water. In practice we also do not know which group an image belongs to during training, which makes the problem harder.

Research scope This week we re-ran three existing methods on Waterbirds, compared our results with the papers, and looked at some of the mistakes they make.

1.2 Literature review

The three methods we study take different approaches. POTER [2] gives more weight to training images that look like the hard cases in a small validation set. DPE [1] trains many small classifiers on top of a frozen model and averages them. AutoBackSwap [3] cuts the bird out of each image and pastes it onto a different background, so the model cannot depend on the background.

1.3 Contributions

- We ran DPE, POTER, AutoBackSwap, and an ERM baseline, and compared the results with the papers.
- We checked the individual test predictions of AutoBackSwap and ERM, and found a possible labeling problem in the dataset and a small bug in the released code.
- We wrote down three ideas for improving DPE, to be tested next week.

2 Related Work

ERM works well on average but is dominated by the majority groups. GroupDRO [4] fixes this by directly improving the worst group, but it needs group labels for every training image. The three methods in this report try to avoid that requirement.

3 Approach

All three methods use a ResNet-50 pretrained on ImageNet.

- **DPE** [1]: first train the model with ERM and freeze it. Then train 15 small classifiers (we call them heads) on balanced subsets of the data, and average their predictions.
- **POTER** [2]: compare the training images with a small validation set whose groups are known, and give higher sampling weight to the images that are similar to the hard groups.
- **AutoBackSwap** [3]: use a detector to find the bird, remove it and fill in the background, then paste the bird onto random backgrounds while training.

4 Experiments

4.1 Dataset and Metrics

We use the official Waterbirds splits [4]: 4,795 training, 1,199 validation, and 5,794 test images. The images fall into four groups: G_1 landbird on land (2,255 test images), G_2 landbird on water (2,255), G_3 waterbird on land (642), and G_4 waterbird on water (642). We report overall accuracy (Acc) and worst-group accuracy (WGA), which is the lowest accuracy among the four groups. The best checkpoint is always chosen on the validation set, never on the test set.

Table 1. Results on Waterbirds (%). Acc denotes overall accuracy and WGA denotes worst-group accuracy. DPE, ERM, and AutoBackSwap are evaluated on the held-out test split. POTER currently reflects validation metrics (epoch 258) for seed 1; held-out test evaluation is pending artifact recovery.

Method and setting	Seeds	Our result		Paper (Test)	
		Acc	WGA	Acc	WGA
DPE, standard	1	91.40	89.00	92.5±0.2	91.0±0.5
DPE, enhanced ERM*	1	93.30	92.70	96.0±0.1	94.1±0.2
POTER (val. checkpoint)	1	91.60†	90.10†	—	90.9±0.3
ERM, no minority	3	68.68±1.00	33.38±1.51	68.4±1.9	33.4±4.4
ERM, with minority	3	93.67±0.23	78.45±2.22	93.6±0.4	74.9±3.3
AutoBackSwap, no minority	3	94.49±0.40	92.33±0.16	95.1±0.3	92.2±0.4
AutoBackSwap, with minority	1	95.44	92.37	95.0±0.5	93.0±0.9

†Validation metrics at epoch 258 (held-out test evaluation pending recovery of test split logs).

4.2 Settings

DPE was run once (seed 0) in two versions: the standard one, and the enhanced one that uses a stronger ERM model (ERM*). POTER was trained once (seed 1) for 300 epochs. AutoBackSwap and ERM were run on a Tesla T4 GPU: seeds 0–2 without minority-group training images, plus one AutoBackSwap run and three ERM runs with them.

4.3 Results and Discussion

Comparison with the papers Table 1 shows our results. DPE reached 89.00% WGA in the standard setting (paper: 91.0±0.5%) and 92.70% in the enhanced setting (paper: 94.1±0.2%), so we are about 2 points below the paper in both. For ERM and AutoBackSwap, our WGA is within two standard deviations of the paper in all four settings. Without minority-group images, AutoBackSwap raises WGA from 33.38% (ERM) to 92.33% (paper: 92.2%). POTER finished training, and its best validation WGA is 90.10% at epoch 258. This is a validation number, so we cannot compare it with the paper's test WGA of 90.9% yet. We still need to recover the test results.

Failure Case Analysis (AutoBackSwap) Recall the four groups: G_1 landbird on land, G_2 landbird on water, G_3 waterbird on land, and G_4 waterbird on water. We looked at the individual test predictions of ERM and AutoBackSwap (seed 0, no minority-group images). ERM makes 1,752 mistakes, mostly on G_2 and G_3 where the background is misleading (accuracy 34.90% and 60.13%). AutoBackSwap makes only 346 mistakes: 75 in G_1, 176 in G_2, 49 in G_3, and 46 in G_4.

We noticed three more things:

- The released code prints the accuracies of G_2 and G_3 with their names swapped in `test_accuracies.txt`. After recomputing from `metadata.csv`, the real worst group is G_2 (92.20%).
- 202 test images are misclassified in all three seeds, and 109 of them are in G_2. So the errors are not random.
- Three species (`102.Western_Wood_Pewee`, `021.Eastern_Towhee`, `088.Western_Meadowlark`; 90 test images) look like landbirds, but `metadata.csv` labels them as waterbirds. AutoBackSwap calls 64 of these 90 images landbirds, which gives 64 of its 346 mistakes and 37 of the 46 mistakes in G_4. These may be label errors in the dataset rather than model errors, but we have not verified this beyond looking at the species names.

Ablation Study (DPE) In the standard setting, the full DPE gets 89.00% WGA. Using random subsets without the covariance term gives 88.20%, and using one fixed subset gives 88.80%. So the full method is better by 0.2–0.8 points, which is a small gap from a single seed. In the enhanced setting all three variants give 92.70–92.80%, so the differences are negligible.

Our Observations and Ideas for DPE We have not yet looked at DPE's failure cases. What follows comes from reading the paper. We have not tested any of these ideas yet and plan to do so next week.

Each of the 15 heads has two prototypes, one for each class, and computes its probabilities from those two only. The final prediction is the simple average of the 15 heads. This made us ask three questions:

1. **Shuffling pairs.** What happens if we mix the prototypes between heads, so that one head's landbird prototype is paired with another head's waterbird prototype? Could this add diversity and improve the result?
2. **Reweighting heads.** All 15 heads currently count equally. If some of them learned the background and are noisy, would giving them lower weights improve accuracy?
3. **More prototypes.** Instead of computing probabilities from just 2 prototypes per head, can we combine more of them in a better way?

5 Conclusion

DPE and AutoBackSwap gave results close to their papers, and POTER still needs its test evaluation. Looking at AutoBackSwap's mistakes, we found that three species may be mislabeled in the dataset, and that they explain about 80% of its G_4 errors. Next week we will finish POTER's test evaluation, look at DPE's failure cases, and try the three ideas above.

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